

a Unified Output Adaptation

Refer to:

"the woman in red holding the umbrella"

Soft probability p

Intervention field u

$$u = 1 - |2p - 1|$$

Hard mask $p = M$

Boundary permission u

$$u = e^{-d(x, \partial M)^2 / (2\sigma^2)}$$

Unified black-box interface
 $(p, u) = \mathcal{A}(O)$



b Reliability-Weighted Inference

Semantic evidence - what to preserve

Regional $\tilde{p}$

Regional $\tilde{u}$

Region graph

anchors t_i

FG anchor

BG anchor

editable

$a_i \uparrow$ as $\tilde{u}_i \downarrow$

Image structure - where to propagate

SLIC superpixels

Region adjacency

Affinity w_{ij}

strong

weak

unique | no learning

$L = D - W$

Sparse SPD solve

$$H = A + \lambda L + \delta I$$

$$Hq^* = At$$

unique | no learning

Regional q^*

c Bounded Pixel Reconstruction

Original p

Gate u

Lifted r

Fusion - pixel-wise gate

$$\alpha = u^\beta, \quad \hat{p} = (1 - \alpha)p + \alpha r$$

Base

Changed

FreeRef

Contour update

original

refined

$$|\hat{p}(x) - p(x)| \leq u(x)^\beta$$

OUTPUT ONLY | NO INTERNAL FEATURES | NO GRADIENTS | NO LEARNED MASK DECODER